

Problem Statement

Project Title

IBM Granite Research Agent – AI-Powered Research Assistant

The Challenge

Researchers, students, and professionals often spend considerable time searching through multiple websites, articles, and research papers to gather accurate information. Existing AI chatbots usually provide generic responses, lose conversational context, and lack organized session management, making research inefficient and time-consuming.

Additionally, switching between different resources and manually organizing research findings reduces productivity and makes it difficult to maintain a structured research workflow.

The Objective

The objective of this project is to develop an **AI-powered Research Agent** using **IBM Granite-4** and **IBM watsonx.ai** that provides intelligent, context-aware, and interactive research assistance.

The system aims to:

- Generate accurate and detailed research responses using IBM Granite models.
- Maintain multi-turn conversations to preserve context during research sessions.
- Securely authenticate users using IBM Identity and Access Management (IAM).
- Save and restore previous research sessions for continuous knowledge exploration.
- Provide a responsive and user-friendly chat interface with quick research prompts.
- Build a scalable architecture that can be extended with Retrieval-Augmented Generation (RAG), document analysis, citation generation, and multi-agent collaboration.

Expected Outcome

The proposed solution delivers a secure, scalable, and intelligent research assistant that improves research productivity, simplifies knowledge discovery, and demonstrates the practical use of **IBM Granite** and **IBM watsonx.ai** for enterprise-ready AI applications.